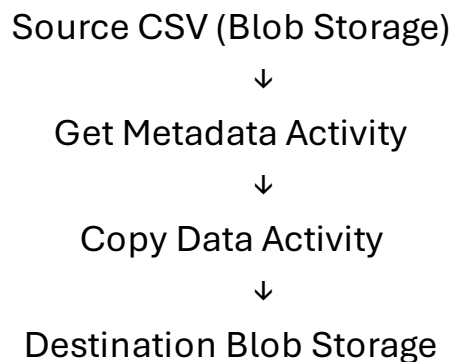


Assignment-4

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

- ★ For every subscription use Azure for students and select the location as Central India(as some are not available in other regions).
- ★ The pipeline flow of the assignment is :



- ★ **Task 1: Resource Group Creation**
- ★ **Task 2: Storage Setup**
- ★ **Task 3: Azure Data Factory Setup**
- ★ **Task 4: Pipeline Development**
- ★ **Task 5: Pipeline Execution and Monitoring**
- ★ **Task 6: IAM Role Assignment**

Task 1: Resource group creation

Exploring the azure portal and the creation of a resource group

- Search for resource groups and create the new resource group by giving name and region.
- Then click on review+create ,resource group will be created.

The screenshot shows the Microsoft Azure portal interface for creating a new resource group. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The breadcrumb trail indicates the path: Home > Resource Manager | Resource groups. The main heading is 'Create a resource group'. Below this, there are three tabs: 'Basics', 'Tags', and 'Review + create'. The 'Basics' tab is currently active. It contains a description of a resource group and three input fields: 'Subscription' (set to 'Azure for Students'), 'Resource group name' (set to 'rg-demo'), and 'Region' (set to '(Asia Pacific) Central India'). At the bottom, there are three buttons: 'Previous', 'Next', and 'Review + create'. The 'Review + create' button is highlighted in blue, indicating it is the next step in the process.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

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CVR COLLEGE OF ENGINEERING I...

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription *

Resource group name *

Region *

Previous Next Review + create

Microsoft Azure

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Home > Resource Manager

Resource Manager | Resource groups

Summarize my costs by service | Export resource groups using Bicep or Terraform | How to manage changes with deployment tools?

Search

+ Create Manage view Refresh Export to CSV Open query Assign tags Add to service group Group by none

Filter for any field... Subscription equals all Location equals all Add filter

Name ↑	Subscription	Location
rg-demo	Azure for Students	Central India

Showing 1 - 1 of 1. Display count: auto

Give feedback

Add or remove favorites by pressing Cmd+Shift+F

Microsoft Azure

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CVR COLLEGE OF ENGINEERING L...

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Automation Link

Basics

Subscription	Azure for Students
Resource group name	rg-demo
Region	Central India

Tags

None

Previous Next Create

- Logged into the Azure portal and navigated to Resource Groups.
- Created a new resource group in Central India region to organize all project resources.
- Verified successful creation of the resource group for further deployment.

Task 2: Storage Setup

- Search for Storage accounts and create the storage account by filling the columns
- Use the resource group same as the resource group created.

The screenshot shows the 'Create a storage account' page in the Microsoft Azure portal. The page is titled 'Create a storage account' and includes a link to 'Learn more about Azure storage accounts'. It is divided into two main sections: 'Project details' and 'Instance details'.

Project details:

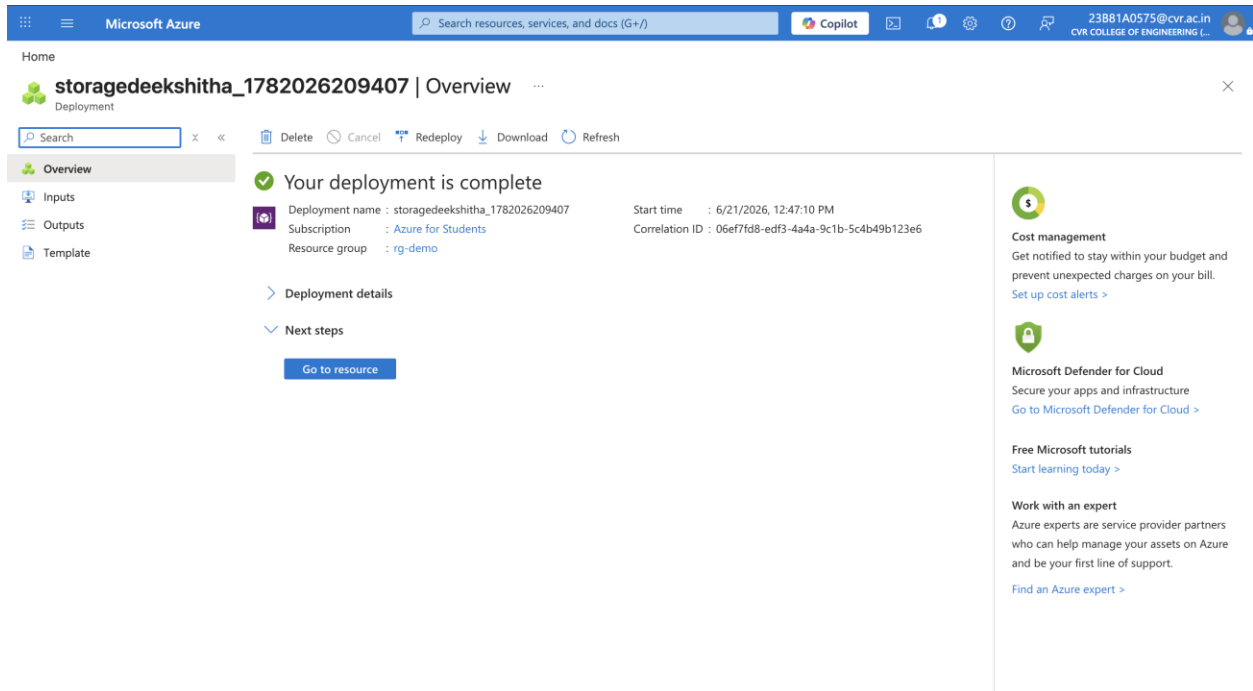
- Subscription ***: Azure for Students
- Resource group ***: rg-demo (with a 'Create new' link below it)

Instance details:

- Storage account name ***: storagedeekshitha
- Region ***: (Asia Pacific) Central India (with a 'Deploy to an Azure Extended Zone' link below it)
- Preferred storage type**: Choose preferred storage type (with an information icon and text: 'This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more')
- Performance ***: ☒ Standard: Recommended for most scenarios (general-purpose v2 account); ☐ Premium: Recommended for scenarios that require low latency.
- Redundancy ***: Locally redundant storage (LRS)

At the bottom, there are three buttons: 'Previous', 'Next', and 'Review + create'. A 'Give feedback' link is also present in the bottom right corner.

After clicking on review+create the storage account will be created.



The above image shows the creation of storage account.

Here we need to create 2 containers for source and destination
in source container it has blob storage (input file).
Destination container is empty at present.

- Creation of containers
- Open the created storage account
- In left panel select data storage and select Container
- Create both the containers

- In source container upload the input csv file

The screenshot displays the Azure Storage portal interface. On the left, a sidebar lists various storage services under the heading 'Data storage'. The 'Containers' option is selected and highlighted. The main area shows the 'New container' dialog box. The 'Name' field is filled with 'source-data' and has a green checkmark. The 'Anonymous access level' dropdown is set to 'Private (no anonymous access)'. A blue information box below the dropdown states: 'The access level is set to private because anonymous access is disabled on this storage account.' At the bottom of the dialog, there is a 'Create' button and a 'Give feedback' link.

storageaccount | Containers

Storage account

Search

+ Add container

Search containers

Showing all 1 items

☐ Name

☐ \$logs

New container

Name *

source-data ✓

Anonymous access level ⓘ

Private (no anonymous access) ▼

i The access level is set to private because anonymous access is disabled on this storage account.

Advanced

Create

Give feedback

Home

source-data

Container

Search

×

«

+ Add Directory

↑ Upload

🔒 Change access level

🔄 Refresh

🗑 Delete

📄 Copy

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

source-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Showing all 0 items

Name

Last modified

Access

No items found

Add or remove favorites by pressing Cnd+Sh1ft+F

Upload blob

1 file(s) selected: Sample - Superstore.csv

Drag and drop files here or [Browse for files](#)

☐ Overwrite if files already exist

Advanced

Upload

Give feedback

Microsoft Azure

Search resources, services, and docs (G+/)

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CVR COLLEGE OF ENGINEERING L...

Home

source-data

Container

Search

+ Add Directory ↑ Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

source-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

	Name	Last modified	Access tier	Blob type	Size	Lease state	
	Sample - Superstore.csv	21/06/2026, 13:01:25	Hot (Inferred)	Block blob	2.18 MiB	Available	...

Add or remove favorites by pressing Cmd+Shift+F

- Created an Azure Storage Account to store project data securely.
- Created two blob containers: source-data and destination-data.
- Uploaded the CSV file into the source container for pipeline processing.

Task 3: ADF

- Create the azure data factory
- And launch the studio
- There we can find the author, monitor and manage options
- In manage, select linked services and create a new linked device.
- In author , select the dataset and create the source and destination datasets.
- Create a new pipeline and drag and drop the metadata

 Microsoft Azure

Home

Create Data Factory

 ...

Basics

 Git configuration Networking Advanced Tags Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Azure for Students

Resource group * ⓘ

rg-demo

[Create new](#)

Instance details

Name * ⓘ

adf-demo-deekshitha

Region * ⓘ

(Asia Pacific) Central India

Version * ⓘ

V2

 We recommend you **upgrade to Fabric Data Factory** for a modern, unified data integration experience. [Try Fabric instead!](#) 

Previous

Next

Review + create

Microsoft Azure

Search resources, services, and docs (G+/I)

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Home

CreateDataFactory-20260621130547 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateDataFactory-20260621130547

Subscription : Azure for Students

Resource group : rg-demo

Start time : 6/21/2026, 1:06:47 PM

Correlation ID : 708dc08d-171f-4865-a0e8-de701d613781

Deployment details

Resource	Type	Status	Operation details
adf-demo-deekshitha	Data factory (V2)	OK	Operation details

Next steps

[Go to resource](#)

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

[Set up cost alerts >](#)

Microsoft Defender for Cloud

Secure your apps and infrastructure

[Go to Microsoft Defender for Cloud >](#)

Free Microsoft tutorials

[Start learning today >](#)

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Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

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Search resources, services, and docs (G+/I)

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Home > CreateDataFactory-20260621130547 | Overview

adf-demo-deekshitha

Data factory (V2)

Summarize the properties of this Data factory (V2).

Show me integration runtime metrics for this Data factory (V2).

+1

Search

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Getting started

Monitoring

Automation

Help

Essentials

Resource group (move) : rg-demo

Status : Succeeded

Location : Central India

Subscription (move) : Azure for Students

Subscription ID : 77ba43eb-82bd-45df-834f-b43e07149250

Type : Data factory (V2)

Getting started : [Quick start](#)

JSON View

Azure Data Factory Studio

Launch studio

Quick Starts

Tutorials

Template Gallery

Training Modules

Monitoring

Microsoft Azure | Data Factory > adf-demo-deekshitha

Search

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Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Azure Data Factory allows you to configure a Git repository with either Azure DevOps or GitHub. Git is a version control system that allows for easier change tracking and collaboration. [Learn more](#)

Set up code repository

Data factory

adf-demo-deekshitha

New

**Ingest**
Copy data at scale once or on a schedule.

**Orchestrate**
Code-free data pipelines.

**Transform data**
Transform your data using data flows.

**Configure SSIS**
Manage & run your SSIS packages in the cloud.

Recent resources

**No items to show**
Your recently opened resources will show up here.

Discover more

Microsoft Azure | Data Factory > adf-demo-deekshitha

Search

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Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Data Factory > Validate all > Publish all

Migrate to Fabric (Preview)

General

Connector upgrade advisor

Factory settings

Connections

Manage

Linked services

Integration runtimes

Microsoft Purview

ADF in Microsoft Fabric

Source control

Git configuration

ARM template

Author

Triggers

Global parameters

Data flow libraries

Security

Credentials

Customer managed key

Outbound rules

Managed private endpoints

Linked services

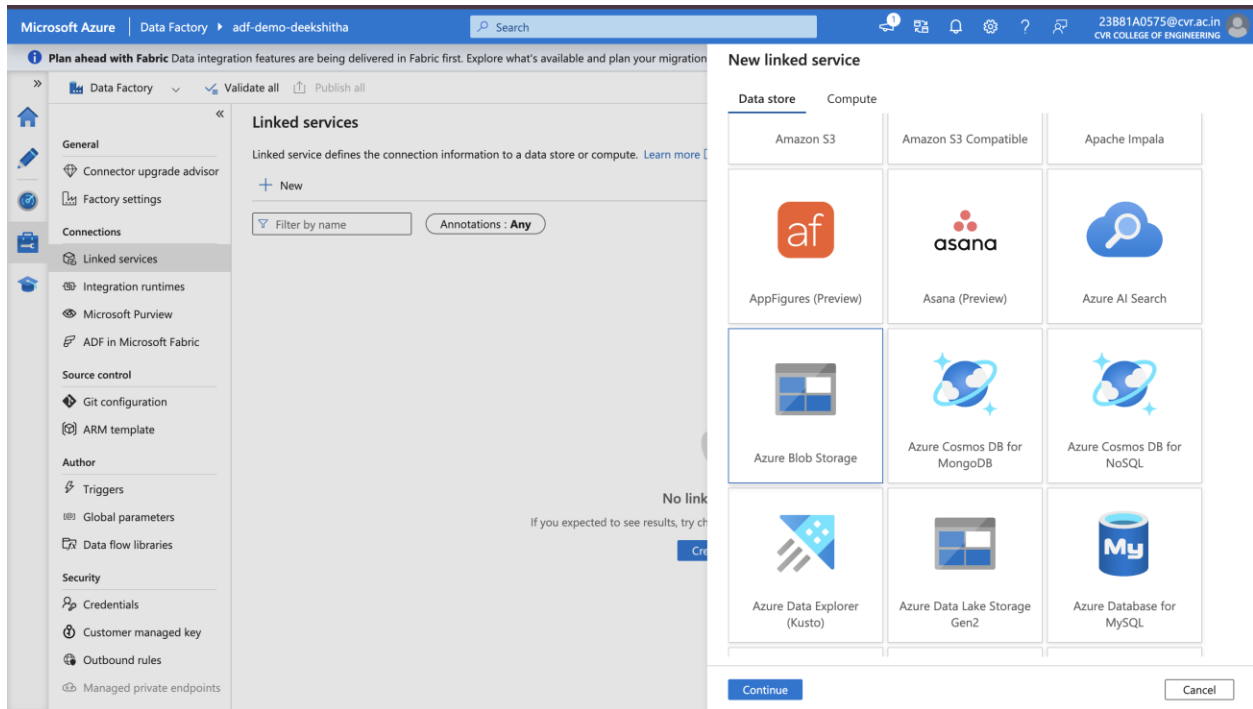
Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name

Annotations : Any

**No linked service to show**
If you expected to see results, try changing your filters or create a new linked services.
[Create linked service](#)



New linked service

Azure Blob Storage [Learn more](#)

Name *

ls_blobstorage

Description

Connect via integration runtime *

AutoResolveIntegrationRuntime

Authentication type

Account key

Connection string

Azure Key Vault

Account selection method

☒ From Azure subscription ☐ Enter manually

Azure subscription

Azure for Students (77ba43eb-82bd-45df-834f-b43e07149250)

Storage account name *

storagedeekshitha

Additional connection properties

+ New

Test connection

☒ To linked service ☐ To file path

Create

Back

Test connection

Cancel

Microsoft Azure | Data Factory > adf-demo-deekshitha

Search

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Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

» Data Factory > Validate all > Publish all

Migrate to Fabric (Preview)

General

Connector upgrade advisor

Factory settings

Connections

Linked services

Integration runtimes

Microsoft Purview

ADF in Microsoft Fabric

Source control

Git configuration

ARM template

Author

Triggers

Global parameters

Data flow libraries

Security

Credentials

Customer managed key

Outbound rules

Managed private endpoints

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name

Annotations : Any

Showing 1 - 1 of 1 items

Name ↑↓	Type ↑↓	Related ↑↓	Annotations ↑↓
ls_blobstorage	Azure Blob Storage	0	

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All

Azure

Database

File

Generic protocol

Amazon S3

Amazon S3 Compatible

Apache Impala

Azure AI Search

Azure Blob Storage

Azure Cosmos DB for MongoDB

Azure Cosmos DB for NoSQL

Azure Data Explorer (Kusto)

Azure Data Lake Storage Gen2

Continue

Cancel

Select format

Choose the format type of your data

 Avro	 Binary	 DelimitedText
 Excel	 JSON	 ORC
 Parquet	 XML	

Continue

Back

Cancel

Set properties

Name

ds_source

Linked service *

ls_blobstorage

File path

source-data

/ Directory

/ sample-superstore.csv

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

Back

Cancel

Set properties

Name

ds_source

Linked service *

ls_blobstorage

File path

source-data

Directory

Sample - Superstore.c

First row as header

Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

Back

Cancel

Microsoft Azure | Data Factory | adf-demo-deekshitha

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1
 - ds_source
- Data flows 0
- Power Query 0

ds_source

DelimitedText ds_source

Connection Schema Parameters

Linked service * ls_blobstorage

File path * source-data / Directory / Sample - Superstore.c

Compression type No compression

Column delimiter ① Comma (,)

Row delimiter ① Default (\r,\n, or \r\n)

Encoding ① Default(UTF-8)

Quote character ① Double quote (")

Escape character ① Backslash (\)

First row as header ① ☒

Properties

General Related

Name * ds_source

Description

Annotations

+ New

Cancel

Set properties

Name

ds_destination

Linked service *

ls_blobstorage

File path

destination-data

/ Directory

/ copied_data.csv

First row as header

Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

Back

Cancel

Microsoft Azure | Data Factory | adf-demo-deekshitha

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 2
 - ds_destination
 - ds_source
- Data flows 0
- Power Query 0

ds_source • ds_destination

DelimitedText
ds_destination

Properties

General Related

Name *
ds_destination

Description

Annotations
+ New

Connection Schema Parameters

Linked service *
ls_blobstorage

File path *
destination-data / Directory / File name

Compression type
No compression

Column delimiter
Comma (,)

Row delimiter
Default (\r,\n, or \r\n)

Encoding
Default(UTF-8)

Quote character
Double quote (")

Escape character
Backslash (\)

First row as header
☒

Cancel

ds_source

ds_destination

pipeline1

Activities

metadata

General

Get Metadata

Validate

Debug

Add trigger

Get Metadata

Get Metadata1

General

Settings

User properties

Dataset *

ds_source

Open

New

Learn more

Field list *

New

Delete

Argument

Exists

Size

Filter by last modified

Start time (UTC)

End time (UTC)

Cancel

Properties

General

Related

Name *

pipeline1

Description

Annotations

New

ds_source

ds_destination

pipeline1

Activities

metadata

General

Get Metadata

Validate

Debug

Add trigger

Get Metadata

check_metadata

Parameters

Variables

Settings

Output

Pipeline run ID

77ed6acc-4bf4-4469-9da3-e5c1cc834fda

Pipeline status

Succeeded

View debug run consumption

All status

Monitor in Azure Metrics

Export to CSV

Showing 1 - 1 of 1 items

Activity name	Activity st...	Activit...	Run start
check_metadata	Succeeded	Get Metadata	6/21/2026, 1:32:5

The screenshot displays the Azure Data Factory (ADF) Studio interface. At the top, the breadcrumb navigation shows 'ds_source' > 'ds_destination' > 'pipeline1'. Below this, the 'Activities' sidebar is visible, showing a search for 'metadata' and a list of activities including 'Get Metadata'. The main canvas shows a pipeline diagram with two activities: 'Get Metadata' and 'check_metadata'. The 'check_metadata' activity is selected, and its output is displayed in a pop-up window. The output is a JSON object containing metadata information. The interface also shows a sidebar with 'Activities' and 'General' sections, and a bottom status bar indicating the pipeline run status as 'Succeeded'.

Output

```
{
  "exists": true,
  "size": 2287806,
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 7
  },
}
```

check_metadata Succeeded Get Metadata 6/21/2026, 1:32:5

- Created an Azure Data Factory instance in the same resource group.
- Explored ADF Studio components such as Author, Monitor, and Manage.
- Configured Linked Service to connect ADF with Azure Blob Storage.

Task 4 & 5 : Pipeline Development and Execution

- Drag and drop copy data
- Connect metadata to copy data
- In copy data put source and destination dataset
- Validate the pipeline
- For output we will have two options debug and trigger
 1. First using debug, if we click on debug button automatically pipeline gets execute.
 2. Using trigger, select trigger now and click ok ,then pipeline line gets executed at the current time only.

The screenshot shows the Azure Data Factory pipeline editor interface. At the top, there are tabs for 'ds_source', 'ds_destination', and 'pipeline1'. Below the tabs, there are buttons for 'Validate', 'Debug', and 'Add trigger'. The main canvas displays a pipeline diagram with two activities: 'Get Metadata' (labeled 'check_metadata') and 'Copy data' (labeled 'Copy data1'). A green arrow indicates the flow from 'Get Metadata' to 'Copy data'. The 'Copy data' activity has a green checkmark, indicating it has been validated. On the right side of the canvas, there are search, add, and zoom controls. Below the canvas, there are tabs for 'Parameters', 'Variables', 'Settings', and 'Output'. The 'Output' tab is selected, showing the pipeline run ID '557dc16d-4132-44bb-afd0-fc64c77e2763' and the status 'Succeeded'. Below this, there is a table showing the execution details of the activities.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	6/21/2026, 2:59:53 PM	16s	AutoResolveIntegrationRuntime (C
check_metadata	✓ Succeeded	Get Metadata	6/21/2026, 2:59:38 PM	14s	AutoResolveIntegrationRuntime (C

Microsoft Azure

Search resources, services, and docs (G+/)

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CVR COLLEGE OF ENGINEERING...

Home > storagedeekshitha | Containers

destination-data

Container

Search

+ Add Directory ↑ Upload 🔒 Change access level ↻ Refresh | 🗑️ Delete 📄 Copy 📄 Paste 🔄 Rename 🔑 Acquire lease 🔑 Break lease 📄 Edit columns

destination-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

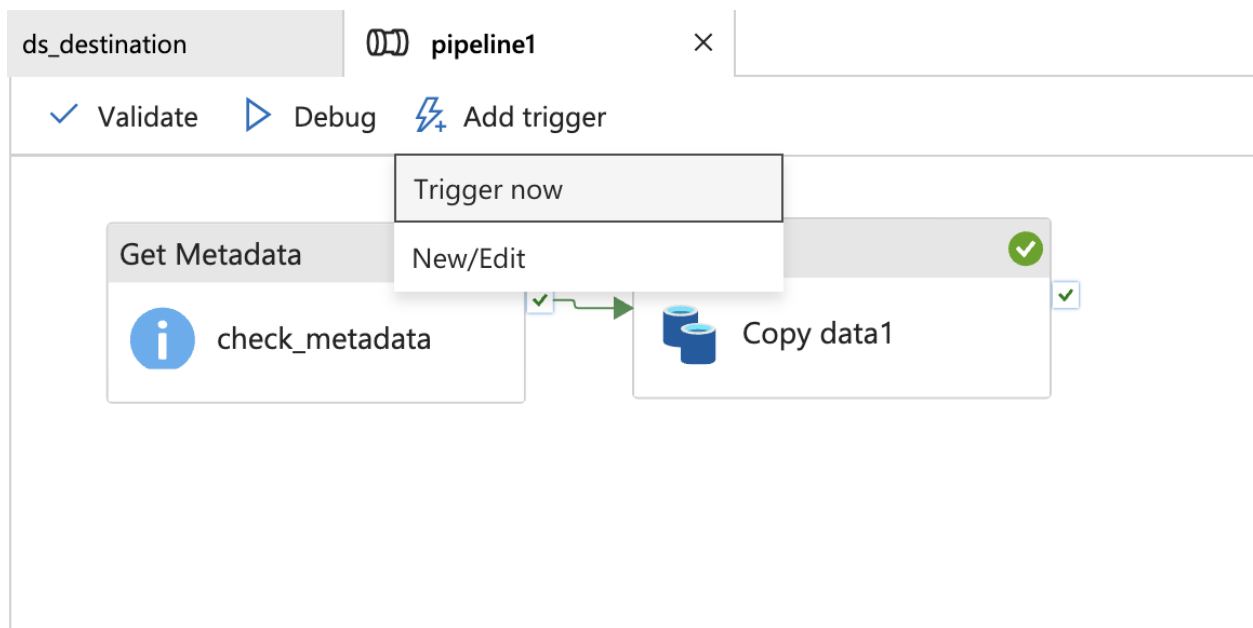
Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	📄 Sample - Superstore.csv	21/06/2026, 15:00:04	Hot (Inferred)	Block blob	2.18 MiB	Available

Add or remove favorites by pressing Cmd+Shift+F

Using trigger we had copied file to destination-data(container) in the storage account.



Pipeline run

Trigger pipeline now using last published configuration.

Parameters

Name	Type	Value
No records found		

OK Cancel

»

«

Home

Dashboard

Runs

Pipeline runs

Trigger runs

Change Data Capture (previ...

Runtimes & sessions

Integration runtimes

Data flow debug

Notifications

Alerts & metrics

Pipeline runs

Triggered Debug

Rerun

Cancel options

Refresh

Edit columns

List

Gantt

Filter by run ID or name

Chennai, Kolkata, Mu... : Last 24 hours

Pipeline name : All

Status : All

Copy filters

Export to CSV

Runs : Latest runs

Triggered by : All

Add filter

×

Showing 1 - 1 items

Last refreshed 0 minutes ago

☐ Pipeline name ↑↓	Run start ↑↓	Run end ↑↓	Duration	Triggered by	Status ↑↓	Run	P
☐ pipeline1	6/21/2026, 3:12:29 PM	6/21/2026, 3:13:12 PM	44s	Manual trigger	✔ Succeeded	Original	

This is done using trigger.

Task 6 : Assigning roles

- Select and open the storage account created.
- On left panel click on Access Control(IAM).
- Select reader and assign members to it.
- And for Contributor do the same as reader

Home > Storage center | Blob Storage > storagedeekshitha | Access Control (IAM)

Add role assignment ...

Role **Members** Conditions Review + assign

Selected role Reader

Assign access to ☒ User, group, or service principal
☐ Managed identity

Members [+ Select members](#)

Name	Object ID	Type	
KATAKAM DEEKSHITHA	3222118d-9be2-4d82-9fec-162cb37a50...	User	

Description

Optional

Review + assign

Previous

Next

[Check access](#) [Role assignments](#) [Roles](#) [Deny assignments](#)

Number of role assignments for this subscription ⓘ

2 4000

Type : All

Role : All

Scope : All scopes

Group by : Role

[All \(2\)](#) [Job function roles \(1\)](#) [Privileged administrator roles \(1\)](#)

☐	Name ↑↓	Type ↑↓	Role ↑↓	Scope ↑↓	Condition ↑↓
▼ Owner (1)					
☐	 KATAKAM DEEKSHITHA 23B81A0575@cvr.ac.in	User	Owner	 Subscription (Inherited)	None
▼ Reader (1)					
☐	 KATAKAM DEEKSHITHA 23B81A0575@cvr.ac.in	User	Reader	 This resource	None

Showing 1 - 2 of 2 results.

For Contributor role i need to go to Privileged administrator roles

Microsoft Azure

Search resources, services, and docs (G+)

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CVR COLLEGE OF ENGINEERING

Home > Storage center | Blob Storage > storagedeekshitha | Access Control (IAM)

Add role assignment

Role

Members

Conditions

Review + assign

Selected role

Contributor

Assign access to

User, group, or service principal

Managed identity

Members

+ Select members

Name	Object ID	Type
KATAKAM DEEKSHITHA	3222118d-9be2-4d82-9fec-162cb37a50...	User

Description

Optional

Review + assign

Previous

Next

Feedback

+ Add

Download role assignments

Edit columns

Refresh

Delete

Feedback

Check access

Role assignments

Roles

Deny assignments

Number of role assignments for this subscription

3

4000

Search by name or email

Type : All

Role : All

Scope : All scopes

Group by : Role

All (3)

Job function roles (1)

Privileged administrator roles (2)

Name	Type	Role	Scope	Condition
Owner (1)				
KD KATAKAM DEEKSHITHA 23881A0575@cvr.ac.in	User	Owner	Subscription (Inherited)	None
Contributor (1)				
KD KATAKAM DEEKSHITHA 23881A0575@cvr.ac.in	User	Contributor	This resource	None

Showing 1 - 2 of 2 results.